

is going to have and their cost. Since software cost is non-recurring (one-time cost only), software development expenditure is recovered after several quantities of a product are sold.

It is the hardware that determines the long term final product cost. So you have to be very focused on the cost sensitivity of your design. Thus, it all comes down to cost versus feature analysis. Inexperienced hardware design engineers are very passionate about incorporating all they can into their design within a short time. However, they should refrain from adding more and more features for expandability and be more cost-sensitive.

So, to get a good overview of what all features should be included, first, make a matrix of different features (such as must have, nice to have, and/or optional features) and the cost involved for each. If the cost constraints do not permit you to have any of those features, then make your system modular wherein a basic system meets the primary objectives of the end product. A few add-on modules can provide the rest. Then you can present it and brainstorm with your marketing/production team (for any addition/changes to the final hardware).

Another thing to consider while designing the hardware is the kind of environment in which it will operate. Ask yourself:

- Is it suitable to work at a normal altitude or at an elevated height above sea level?
- What is going to be its operating temperature range?
- Should it be fitted in air-tight or weatherproof enclosures?
- What is going to be the air-circulation or cooling method?

Enclosures are important as they give a first general impression to users regarding a product's reliability and characteristics such as display and type of user interface. The user interface cannot be overlooked as it plays an important role in overall functioning of the product. UI is the user's window through which the user accesses your product. Ignoring certain aspects and straightway jumping to hardware design can lead to severe consequences later on.

Whether you plan to sell your

Things to remember

1. To check whether the temperature rise is permissible or not, study the IEC standards associated with tolerable temperature rise after the equipment is powered on for a long time.
2. The efficiency and component cost of a power supply are directly proportional. An efficient power supply will greatly reduce heat dissipation and cost.
3. For an efficient power supply, you need inductors that have very low DC resistance, capacitors with low ESR, and very low leakage diodes.

product in India or abroad, appropriate standards and certifications are required, such as:

India: Bureau of Indian Standards (BIS)

US or Canada: Underwriters Laboratories (UL) (CSA approval for Canada)

Europe: CE mark or EN standard

In most cases, these standards are very similar. But they should meet specific criteria regarding EMI/EMC requirements and different tolerance levels, etc. So read the applicable standards and then design the hardware.

Besides these, there are several requisite approvals for other countries as well, which must be complied. Else, once your design is ready, it will struggle to pass the safety regulations without appropriate compliances.

These are the basic things one should always begin with.

Now hardware design begins

After recognising all of the above, the initial documentation is prepared as per the design life cycle you are following. Make high-level and low-level design diagrams. List down design constraints, tolerances, and interface diagrams. Then the actual hardware design can begin. Start with the selection of relevant electronic components. Try to have alternate parts for most of the important components with pin compatibility. Check for obsolescence and life expectancy of your product. You also have to write the test cases and define their acceptance criteria. Focus on stating the grounds on which you are going to

assume the hardware is perfect. Get it approved from certifying agencies and assure that your test criteria meet the compliances that you are aiming for.

Having done all this, the major challenge is fitting the PCBs into the enclosures. At this stage, several questions arise:

- How will the internal wiring be done?
- Where are the external wires going to or where they come from?
- How is the product going to be installed on the field?

Again, inexperienced engineers tend to forget these things. Because internal wiring may take up a lot of space within the product enclosure, one has to carefully think about how it should be carried out, avoiding the interference.

Things to consider before wiring:

- Size and capacity of the screw terminals
- The current requirement of the screw terminals
- Should it be pluggable?
- D-Sub connectors or pluggable screw terminals for external connection?

After freezing the connector/screw terminal locations, you will get an idea regarding the correct positioning of electronic components on the PCB. It cannot be reversed or other way around. Making space for silicon components/functional boards first and then thinking of how it will be placed into the enclosure may be disastrous. Be careful about the pluggable part of the connector so it does not foul with other parts of hardware, enclosure etc. Such things are often forgotten.

Planning for optimal power consumption

Whether the product/device is battery or mains power operated, ensure to take temperature ranges, heat dissipation and power supply consumption into account.

A microcontroller (MCU) or a microprocessor (MPU) consumes the energy according to its throughput. Power consumption is directly proportional to the controller performance in most cases.

More clock speed equals an execution speed of million instructions per second. Again, in the case of multi-